# A Leakage-Safe Temporal-Evidence Pipeline for Assessing Network Impact on Manufacturing Efficiency in Smart Factories: A Case Study

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***Abstract*—The paradigm of Industry 5.0 and smart manufacturing envisions pervasive integration of 6G Ultra-Reliable Low-Latency Communication (URLLC) to drive real-time closed-loop control and autonomous cyber-physical coordination. However, applied machine learning (ML) models evaluating network telemetry frequently suffer from methodological vulnerabilities, including temporal data leakage, target contamination, and unvalidated synthetic benchmark assumptions. In this paper, we propose an end-to-end, leakage-safe temporal-evidence pipeline for auditing network telemetry and quantifying its causal and predictive influence on industrial manufacturing efficiency. Through comprehensive Kolmogorov-Smirnov (KS) goodness-of-fit testing across nine industrial features, we reveal that the studied industrial telemetry dataset ($N = 100,000$ observations across 50 machines over 68.4 days) shows no evidence against continuous uniform generation; no pairwise feature correlation survives Bonferroni correction across all 36 pairs (maximum $|r| = 0.0075$ between Vibration and Network Latency, uncorrected $p = 0.0170$, adjusted $p = 0.6127$), demonstrating absence of linear dependence. While a naive segmented regression search suggests an Active-mode latency breakpoint at 20.7 ms ($F = 4.3127$, naive asymptotic $p = 0.0378$), a rigorous supremum-$F$ permutation test ($N = 2,000$, seed=42) reveals $p = 0.1640$ (FDR $q = 0.5100$), establishing the threshold as exploratory rather than confirmatory. Under the permutation null distribution, the naive test rejects the null at nominal $\alpha = 0.05$ in $20.45\%$ of realizations (409/2,000), proving a more than fourfold inflation of type-I error. Utilizing leak-free temporal and machine-grouped validation splits, we benchmark Random Forest and LightGBM models against naive baselines: network telemetry achieves near-chance classification discrimination (ROC-AUC 0.4738–0.5073) for composite efficiency, failing to outperform random guessing. Our findings highlight the critical necessity of rigorous data audits and leak-free temporal benchmarking in smart factory research.**

***Index Terms*—Industrial IoT, 6G URLLC, Predictive Maintenance, Temporal Leakage, Statistical Audit, Changepoint Detection, Machine Learning in Manufacturing.**

## I. Introduction

### A. The Promise: 6G URLLC in Industry 4.0/5.0

The convergence of advanced telecommunications and industrial automation forms the core infrastructure of Industry 4.0 and 5.0. Future Sixth-Generation (6G) wireless communication networks promise to deliver sub-millisecond round-trip latencies, extreme jitter bounds, and block reliability rates exceeding $99.99999\%$ ($10^{-7}$ block error rate) [1], [2]. In mission-critical cyber-physical production systems (CPPS), such Ultra-Reliable Low-Latency Communication (URLLC) capabilities are designed to underpin closed-loop robotic cell collaboration, distributed programmable logic controllers (PLCs), edge-native computer vision, and real-time automated guided vehicle (AGV) routing [3], [4].

Standardization bodies, notably 3GPP in Technical Specification TS 22.104 [5], have articulated precise communication service requirements for vertical domains, specifying end-to-end latency tolerances and message transfer intervals across diverse motion control and machine interaction operational scenarios (e.g., Clause 5.1.1 and Table 5.1.1-1). In parallel, smart factory operators increasingly ingest wireless telemetry (e.g., latency, jitter, packet loss) alongside operational technology (OT) machine sensors (e.g., temperature, vibration, power consumption) into predictive machine learning pipelines to forecast production throughput and avoid costly operational downtime.

### B. The Problem: Data Leakage and Optimistic Evaluation

Despite significant investments in smart manufacturing analytics, a profound reproducibility crisis affects applied predictive modeling [6]. Methodological surveys across data mining and applied machine learning demonstrate that published predictive architectures frequently report inflated classification performance due to subtle forms of data leakage [6], [7]. Specifically, two vulnerabilities are rampant:

1) **Temporal Autocorrelation and Group Contamination**: Standard random $K$-fold cross-validation mixes temporally adjacent observations across train and test partitions. When machines operate under continuous autoregressive processes, random splits leak temporal dependencies, causing models to memorize specific time-series trajectories rather than generalizing to unseen temporal horizons [8].
2) **Target Contamination / Proxy Feature Leakage**: Target labels such as "Overall Equipment Efficiency" (OEE) or categorical "Operational Efficiency" are frequently derived as deterministic or thresholded composite functions of raw operational features (e.g., error counts, defect counts, production rates). Including raw or weakly masked components of the composite label as predictive covariates creates circular tautologies where ML models merely invert the labeling heuristic rather than learning physical degradation phenomena.

TABLE I
RESEARCH QUESTIONS ADDRESSED IN THIS STUDY

| ID | Core Investigation |
|---|---|
| **RQ1** | What is the true statistical generative distribution of the industrial telemetry data, and does it exhibit authentic cross-feature physical correlations? |
| **RQ2** | Do network telemetry metrics (latency, packet loss) demonstrate temporal stationarity and causal precedence over production speed? |
| **RQ3** | Are candidate structural breakpoints in network latency statistically robust when controlling for data-chosen threshold search? |
| **RQ4** | Do unsupervised network telemetry clusters reflect distinct physical operational regimes or continuous geometric partitions? |
| **RQ5** | What is the true predictive power of network telemetry for manufacturing efficiency under strictly leak-free temporal and machine-grouped validation? |

### C. The Reality Gap: Synthetic Uniform Benchmarks

Compounding these evaluation failures, researchers and practitioners often benchmark complex predictive algorithms on public or semi-synthetic industrial datasets without verifying the underlying generative process. When synthetic benchmarks are produced via independent pseudorandom uniform sampling, physical interactions—such as network congestion directly degrading production cadence—vanish.

Treating such datasets as empirical industrial telemetry leads to erroneous conclusions: data-mining algorithms may detect spurious statistical associations, cluster continuous noise into artificial operational "regimes," and report deceptive structural breakpoints.

### D. Research Questions

To address these pervasive challenges, this investigation establishes a systematic, leakage-safe empirical framework guided by five core Research Questions (RQs), summarized in Table I.

### E. Summary of Contributions

In this paper, we present the design, mathematical formalization, and empirical execution of a leakage-safe temporal-evidence pipeline applied to a smart factory telemetry benchmark. Our primary contributions are:

1) **Rigorous Generative and Label Auditing**: We conduct formal Kolmogorov-Smirnov (KS) goodness-of-fit testing across nine continuous telemetry variables and analyze 36 pairwise correlations. We show that the dataset exhibits no evidence against independent uniform generation, and formally audit the deterministic composite logic governing efficiency labels.
2) **Methodological Finding on Data-Chosen Breakpoints**: We demonstrate that testing data-chosen structural breakpoints using standard naive asymptotic $F$-distributions yields substantial type-I error inflation ($F = 4.3127$, naive asymptotic $p = 0.0378$). Under the permutation null of independence, the naive test rejects at nominal $\alpha = 0.05$ in $20.45\%$ of realizations (409/2,000). We implement a vectorized supremum-$F$ permutation test ($N = 2,000$) establishing that the Active-mode threshold ($\tau = 20.69$ ms) is exploratory ($p = 0.1640$, FDR $q = 0.5100$).
3) **Leakage-Safe Temporal Benchmarking**: We evaluate Random Forest and LightGBM across temporal and machine-grouped partitions, strictly isolating contemporaneous label components to prevent circular data leakage, and demonstrate that network telemetry yields near-chance classification (ROC-AUC 0.4738–0.5073).
4) **Open-Source Verifiable Pipeline**: We release the entire experimental codebase, automated data auditing suites, interactive Streamlit operational dashboard, and reproducibility scripts as an open-source framework for the industrial informatics community.[1]

## II. RELATED WORK

The intersection of wireless communications, industrial telemetry, and predictive modeling has attracted widespread investigation. Table II summarizes representative studies examining network QoS, control performance, and methodological validation in industrial environments.

### A. Network Telemetry in Smart Factories

The integration of 5G and nascent 6G private cellular networks provides industrial environments with fine-grained telemetry tracking packet latency, radio signal strength indicators (RSSI), and transmission error rates [1]. Several recent architectures explore localized breakout mechanisms at the gNodeB (gNB) to minimize propagation delays for robotic workcells [2], [3]. Communication-theoretic frameworks characterize fundamental trade-offs between throughput, latency, and reliability in network slicing [13]. In experimental settings, redundant transmission paths and diversity schemes have been shown to stabilize latency under wireless channel degradation [9], [10]. Furthermore, physical closed-loop control studies confirm that uncompensated network transport delays can impair plant stability [11]. However, translating these microscopic networking dynamics into reliable macroscopic predictive signals remains challenging when operational telemetry exhibits low signal-to-noise ratios.

### B. Predictive Modeling and Methodological Pitfalls

In predictive modeling on tabular sensor telemetry, ensemble tree methods such as Random Forests [14] and gradient boosted decision trees (e.g., LightGBM [15], XGBoost [16]) are widely adopted for their handling of heterogeneous features. However, as documented by Kaufman *et al.* [7] and Kapoor and Narayanan [6], predictive pipelines in applied informatics frequently suffer from proxy feature leakage. When machine efficiency is defined via composite scoring rules that directly aggregate individual telemetry streams, standard cross-validation fails to detect that the classifier is simply reproducing the arithmetic definition of the target. In this study, we integrate formal distribution audits, non-parametric

[1] https://github.com/Atharva110409/smart-factory-6g-network

TABLE II
SURVEY OF RELATED STUDIES IN INDUSTRIAL COMMUNICATION, TELEMETRY, AND PREDICTIVE VALIDATION

| Reference | Title Returned by DOI | Domain / Focus | Data Origin | Key Empirical / Methodological Finding |
|---|---|---|---|---|
| Cena *et al.* [9] (DOI: 10.1109/TII.2020.2979759) | Redundancy and Diversity in Wireless Networks to Support Mobile Industrial Applications in Industry 4.0 | Industrial Wireless Networks | Physical Factory Testbed | Seamless packet replication across Wi-Fi channels bounds latency and mitigates shadowing for automated guided vehicles. |
| Zunino *et al.* [10] (DOI: 10.1109/TII.2023.3271708) | Adaptive Seamless Redundancy to Achieve Highly-Dependable MQTT Communication | IIoT Messaging QoS | Emulation / Testbed Logs | Adaptive path selection dynamically minimizes transmission delays for critical MQTT industrial telemetry streams. |
| Ferrari *et al.* [11] (DOI: 10.1109/TII.2010.2089059) | Wireless Model-Based Predictive Networked Control System Over Cooperative Wireless Network | Networked Control Systems | Closed-Loop Testbed | Empirical packet latency fluctuations directly degrade predictive control loop stability unless explicitly compensated. |
| Vitturi *et al.* [12] (DOI: 10.1109/JPROC.2019.2913443) | Industrial Communication Systems and Their Future Challenges: Next-Generation Ethernet, IIoT, and 5G | Industrial Communication Survey | Industry Standards / OT | Synthesizes deterministic QoS bounds, sub-millisecond target latencies, and migration paths toward 5G URLLC in manufacturing. |
| Popovski *et al.* [13] (DOI: 10.1109/ACCESS.2018.2872781) | 5G Wireless Network Slicing for eMBB, URLLC, and mMTC: A Communication-Theoretic View | Wireless Slicing Theory | Analytical / Sim | Analyzes fundamental trade-offs between throughput, packet delay, and reliability across heterogeneous slices. |
| Kapoor & Narayanan [6](DOI: 10.1016/j.patter.2023.100804) | Leakage and the Reproducibility Crisis in Machine-Learning-Based Science | ML Reproducibility | Meta-Analysis ($N = 329$ papers) | Identifies data leakage as a widespread driver of irreproducible claims across applied scientific machine learning. |
| **This Work** | — | **Industrial Telemetry Audit** | **Industrial Benchmark** | **Integrates distribution audits, sup-$F$ permutation testing, and leak-free temporal modeling to expose false signals.** |

permutation breakpoint testing, and leak-free temporal validation into a unified smart factory telemetry pipeline.

## III. METHODOLOGY AND DATASET AUDIT

### A. Pipeline Architecture Overview

The end-to-end analytical framework developed in this study is illustrated in Fig. 1. The pipeline enforces strict unidirectional data flow across four decoupled modules: (1) Generative & Distributional Audit, (2) Temporal & Structural Changepoint Analysis, (3) Unsupervised Continuum Profiling, and (4) Leakage-Safe Predictive Modeling.

### B. Dataset Provenance and Statistical Distribution Audit

The case study analyzes the *Thales Group Manufacturing Telemetry* dataset released on Kaggle, capturing smart factory operational records. The dataset comprises $N = 100,000$ telemetry rows recorded across 50 distinct industrial machine identifiers over a 68.4-day span (from 2025-01-01 00:00:00 to 2025-03-10 10:39:00). While dataset-wide timestamps advance regularly, per-machine observations arrive irregularly (median inter-sample arrival time of approximately 34 minutes per machine, with an interquartile range of 15 to 68 minutes). Each record logs timestamp, machine identifier ($M \in [1, 50]$), operating mode ($S \in \{\text{Active}, \text{Idle}, \text{Maintenance}\}$), two wireless network telemetry features, and six operational/environmental sensor variables.

To address **RQ1**, we test whether each continuous telemetry variable $X$ exhibits significant departure from a continuous uniform distribution $\mathcal{U}(a, b)$, where $a = \min(X)$ and $b = \max(X)$. We apply the two-sided one-sample Kolmogorov-Smirnov (KS) test:

$$D_N = \sup_{x \in [a,b]} |F_N(x) - F_{\mathcal{U}}(x)|, \tag{1}$$

where $F_N(x) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}_{x_i \le x}$ is the empirical cumulative distribution function (ECDF), and $F_{\mathcal{U}}(x) = \frac{x-a}{b-a}$ is the theoretical uniform CDF.

Furthermore, to test for multivariate linear dependencies, we compute the Pearson correlation matrix $\mathbf{R} \in \mathbb{R}^{9\times 9}$ across all $\binom{9}{2} = 36$ distinct feature pairs and apply the family-wise Bonferroni correction:

$$p_{\text{adj}} = \min\left(1.0,\ 36 \times p_{\text{uncorr}}\right). \tag{2}$$

### C. Audited Label Rule Construction and Imbalance Ratio

The operational target in the benchmark is categorical *Operational Efficiency*, categorized into three ordinal levels: $\mathcal{Y} \in \{\text{Low}, \text{Medium}, \text{High}\}$. Auditing the ground truth against operational telemetry reveals that $\mathcal{Y}$ is governed by a deterministic composite decision boundary ($99.998\%$ exact match across all $100,000$ records):

$$\mathcal{Y} = \begin{cases} \text{Low}, & \text{if } E > 5.0 \ \vee \ P \le 200, \\ \text{High}, & \text{else if } P > 400 \ \wedge \ E \le 2.0, \\ \text{Medium}, & \text{otherwise}, \end{cases} \tag{3}$$

where $E$ denotes Error Rate (%) and $P$ denotes Production Speed (units/hr).

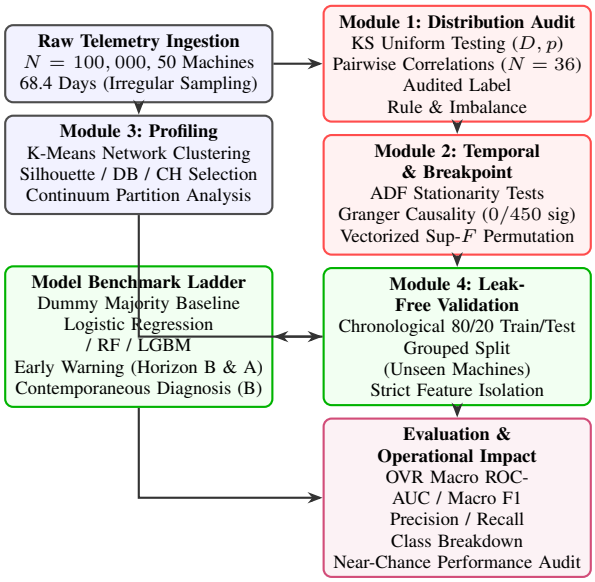



Fig. 1. End-to-End Leakage-Safe Temporal-Evidence Pipeline for Smart Factory Telemetry.

$$\begin{aligned} P(\text{Low}) &= 1 - (1/3)(2/3) = 7/9 \approx 77.78\%, \\ P(\text{High}) &= (2/9)(2/15) = 4/135 \approx 2.96\%, \\ P(\text{Medium}) &= 1 - 7/9 - 4/135 = 26/135 \approx 19.26\%. \end{aligned} \tag{4}$$

Recomputing the exact empirical class frequencies from the CSV confirms: Low Efficiency ($N_{\text{Low}} = 77,825$, $77.825\%$), Medium Efficiency ($N_{\text{Med}} = 19,189$, $19.189\%$), and High Efficiency ($N_{\text{High}} = 2,986$, $2.986\%$). To quantify class asymmetry, we compute the empirical class imbalance ratio between the majority and minority classes:

$$\text{Imbalance Ratio} = \frac{N_{\text{Low}}}{N_{\text{High}}} = \frac{77,825}{2,986} \approx 26.06 : 1. \tag{5}$$

Over $77.8\%$ of operational states fall into the degraded "Low" efficiency regime, while pristine "High" efficiency represents under $3\%$ of operations.

### D. Temporal Stationarity and Causal Precedence

To address **RQ2**, we evaluate the time-series dynamics of network latency ($L_t$) and production speed ($P_t$). We test for unit roots using the Augmented Dickey-Fuller (ADF) regression implemented via `statsmodels` [17], [18]:

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \sum_{j=1}^{p} \delta_j \Delta Y_{t-j} + \varepsilon_t. \tag{6}$$

Rejecting the null hypothesis $H_0 : \gamma = 0$ confirms weak stationarity. ADF tests are conducted on both the continuous network latency and production speed series across all machines.

Because per-machine telemetry sampling is irregular, autoregressive lags represent sequential row lags per machine rather than fixed elapsed minutes. We evaluate Granger causality across all 50 individual machines and operational modes for row lags $k \in \{1, 2, 3\}$ (450 distinct tests) [19]. We estimate the bivariate autoregressive model:

$$P_t = c + \sum_{i=1}^{k} a_i P_{t-i} + \sum_{i=1}^{k} b_i L_{t-i} + u_t, \tag{7}$$

and test the joint restriction $H_0 : b_1 = \cdots = b_k = 0$ using an $F$-test. To guard against inflated false discovery rates across multiple tested configurations, we apply the Benjamini-Hochberg (BH) procedure [20], enforcing an FDR threshold of $q = 0.05$.

### E. Structural Breakpoint Detection: Naive $F$ vs. Sup-$F$ Permutation

To address **RQ3**, we examine whether network latency exhibits a sharp threshold beyond which production speed collapses. A segmented linear regression model parameterizes this relationship across candidate breakpoint $\tau \in [\tau_{\min}, \tau_{\max}]$:

$$P_t = \beta_0 + \beta_1 L_t + \beta_2 (L_t - \tau) \cdot \mathbf{1}_{\{L_t > \tau\}} + \varepsilon_t. \tag{8}$$

Testing the presence of a structural break evaluates the null hypothesis $H_0 : \beta_2 = 0$ against $H_1 : \beta_2 \neq 0$. Candidate thresholds are scanned across the 10th to 90th empirical percentiles of the latency distribution (81 quantile points). In standard software implementations, an exhaustive grid search selects $\hat{\tau} = \arg\max_\tau F(\tau)$, where $F(\tau)$ is the Chow-type $F$-statistic comparing the segmented model against a simple linear baseline.

**Methodological Hazard**: As proved by Andrews [21] and Hansen [22], when the threshold parameter $\tau$ is unknown under the null hypothesis (nuisance parameter problem), the test statistic $\sup_\tau F(\tau)$ does *not* follow a standard Fisher-Snedecor $F_{1,N-k}$ distribution. Selecting the maximal $F$-statistic across a dense grid causes the naive asymptotic $p$-value to be substantially anti-conservative.

To obtain an exact, finite-sample valid test of independence between latency and production speed without relying on asymptotic approximations, we implement a vectorized supremum-$F$ permutation test with $B = 2,000$ iterations (seed=42 in `src/audit_uniformity.py`). Utilizing the Frisch-Waugh-Lovell (FWL) theorem, we project out the linear nuisance subspace $\mathbf{X}_0 = [\mathbf{1}, \mathbf{L}]$:

$$\mathbf{M}_0 = \mathbf{I} - \mathbf{X}_0(\mathbf{X}_0^T\mathbf{X}_0)^{-1}\mathbf{X}_0^T. \tag{9}$$

For each candidate $\tau$, the orthogonalized segmented regressor is $\tilde{\mathbf{z}}_\tau = \mathbf{M}_0(\mathbf{L} - \tau)_+$. Under the permutation null of independence, for any randomly permuted response vector $\mathbf{P}^{(b)}$, the permuted response is residualized via $\tilde{\mathbf{P}}^{(b)} = \mathbf{M}_0\mathbf{P}^{(b)}$. Because $\mathbf{M}_0$ is symmetric and idempotent with $\mathbf{M}_0\tilde{\mathbf{z}}_\tau = \tilde{\mathbf{z}}_\tau$, the numerator of the test statistic simplifies to a vectorized dot product:

$$F^{(b)}(\tau) = \frac{(\tilde{\mathbf{z}}_\tau^T\tilde{\mathbf{P}}^{(b)})^2/(\tilde{\mathbf{z}}_\tau^T\tilde{\mathbf{z}}_\tau)}{\hat{\sigma}^2_{(b)}(\tau)}. \tag{10}$$

The empirical supremum permutation $p$-value is computed as:

$$p_{\text{perm}} = \frac{1}{B}\sum_{b=1}^{B}\mathbf{1}_{\{\sup_\tau F^{(b)}(\tau) \geq \sup_\tau F_{\text{obs}}(\tau)\}}. \tag{11}$$

To estimate parameter uncertainty for $\hat{\tau}$ in pooled regression while accounting for machine-level clustering, we implement a machine-block bootstrap ($B = 1,000$ resamples). Resampling intact machine time-series with replacement preserves within-machine serial correlation and operational heterogeneity, yielding empirical confidence intervals for $\hat{\tau}$.

### F. Leakage-Safe Predictive Modeling Framework

To answer **RQ5**, we formulate the prediction of operational efficiency $\mathcal{Y}_t$ as a supervised multiclass classification task implemented using `scikit-learn` [23] and LightGBM [15].

**Feature Engineering and Target Isolation**: To prevent proxy feature leakage, contemporaneous Error Rate and Production Speed are strictly excluded from all feature matrices. The predictive feature set (32 features total) includes: raw operational telemetry (Temperature, Vibration, Power Consumption, Maintenance Score, Defect Rate); wireless network telemetry (Network Latency, Packet Loss); sequential lag features (Lags 1, 2, and 3 for Latency and Packet Loss); and rolling window statistics (mean, standard deviation, min, max over 5-step and 15-step windows).

**Architectures and Hyperparameters**: We benchmark four supervised models against a Dummy Majority baseline: (1) *Logistic Regression*: $L_2$ regularization ($C = 1.0$), 'class_weight='balanced'', maximum iterations 1,000 with 'StandardScaler'; (2) *Random Forest*: 200 trees, maximum depth 12, minimum samples split 5, 'class_weight='balanced''; and (3) *LightGBM*: 200 boosting iterations, learning rate 0.05, maximum depth 6, 31 leaves, 'class_weight='balanced''.

**Formulations and Validation Splits**: We evaluate four experimental configurations: (1) *Model A (Early Warning, Horizon B)*: predicts future efficiency $\mathcal{Y}_{t+H}$ at physical time horizon $H = 30$ min $\pm$ 10 min ($N = 23,587$); (2) *Model A (Early Warning, Horizon A)*: sensitivity check using a 1-step row shift ($N = 99,950$); (3) *Model B (Diagnosis)*: contemporaneous classification at time $t$ strictly isolating raw label components; and (4) *Model A (Unseen Machines 41–50, Horizon B)*: trained on Machines 1–40 and tested on held-out Machines 41–50 under Horizon B ($N = 20,016$). Validation enforces an 80/20 chronological split ($t_{\text{train}} < t_{\text{test}}$) as well as grouped holdout splits by machine identifier. Feature attribution is computed via SHAP [24].

TABLE III
KOLMOGOROV-SMIRNOV GOODNESS-OF-FIT TESTS AGAINST UNIFORM DISTRIBUTION $\mathcal{U}(a, b)$ ($N = 100,000$)*

| Variable | Bounds $[a, b]$ | KS Stat $D$ | Raw $p$ | Bonf. $p$ |
|---|---|---|---|---|
| Network Latency (ms) | $[1.000, 50.000]$ | 0.003059 | 0.3061 | 1.0000 |
| Packet Loss (%) | $[0.000, 5.000]$ | 0.003540 | 0.1627 | 1.0000 |
| Production Speed (u/h) | $[50.000, 499.997]$ | 0.004756 | 0.0216 | 0.1947 |
| Error Rate (%) | $[0.000, 15.000]$ | 0.002063 | 0.7873 | 1.0000 |
| Defect Rate (%) | $[0.000, 10.000]$ | 0.002680 | 0.4683 | 1.0000 |
| Temperature (°C) | $[30.000, 89.999]$ | 0.001985 | 0.8248 | 1.0000 |
| Vibration (Hz) | $[0.100, 5.000]$ | 0.001902 | 0.8617 | 1.0000 |
| Power Cons. (kW) | $[1.500, 10.000]$ | 0.002114 | 0.7622 | 1.0000 |
| Maintenance Score | $[0.000, 1.000]$ | 0.002850 | 0.3903 | 1.0000 |

*KS bounds were estimated from data; $p$-values are conservative (Lilliefors-type correction not applied).

## IV. EMPIRICAL RESULTS

### A. Statistical Distribution and Multicollinearity Audit (RQ1)

Table III details the results of the Kolmogorov-Smirnov goodness-of-fit testing against continuous uniform distributions across all nine numeric features in the dataset, generated by `src/audit_uniformity.py` and archived in `reports/ks_uniformity_tests.csv`.

The empirical findings show clear concordance across tests. For eight of the nine variables, raw $p$-values exceed 0.16, showing no evidence against uniform distributions. For Production Speed, the test yields $D = 0.004756$ with raw $p = 0.0216$. However, when correcting for multiple testing across all nine variables, the Bonferroni-adjusted $p$-value is $p_{\text{adj}} = 0.1947$ (and across the five core operational variables, $p_{\text{adj}} = 0.1082$); neither reaches the standard significance threshold $\alpha = 0.05$. Thus, all nine features show *no statistical evidence against independent uniform generation*.

Furthermore, evaluation of the pairwise correlation matrix $\mathbf{R} \in \mathbb{R}^{9\times9}$ (archived in `reports/correlation_matrix.csv`) reveals that all 36 pairwise Pearson correlation coefficients are bounded in $[-0.0071, +0.0075]$. The single largest absolute correlation observed across the entire dataset is between Vibration and Network Latency ($r = 0.007546$, raw uncorrected $p \approx 0.0170$). After family-wise Bonferroni adjustment across all 36 pairs, the adjusted $p$-value is $p_{\text{adj}} = \min(1.0, 36 \times 0.017019) = 0.6127$. Not a single correlation survives multiple testing correction. These

TABLE IV
GRANGER CAUSALITY ANALYSIS: NETWORK LATENCY → PRODUCTION SPEED (450 CONFIGURATIONS TESTED)

| Operation Mode | Tests | Raw $p < 0.05$ | FDR $q < 0.05$ | Significant? |
|---|---|---|---|---|
| Active | 150 | 7 (4.7%) | 0 (0.0%) | No |
| Idle | 150 | 9 (6.0%) | 0 (0.0%) | No |
| Maintenance | 150 | 7 (4.7%) | 0 (0.0%) | No |
| **Total / Overall** | **450** | **23 (5.1%)** | **0 (0.0%)** | **No (0 / 450)** |

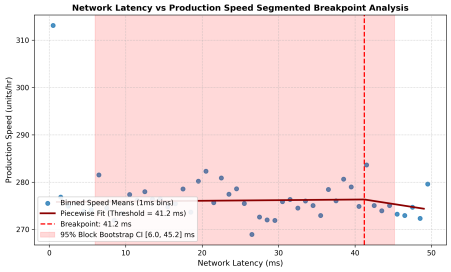



Fig. 2. Segmented piecewise linear regression fit of network latency versus production speed for pooled telemetry across all 50 machines ($N = 100,000$). Global nominal breakpoint at $\hat{\tau} = 41.22$ ms ($F = 1.0532$, naive $p = 0.3048$, sup-$F$ perm $p_{\text{perm}} = 0.7530$, FDR $q = 0.7530$) yields an unreliably wide 95% block bootstrap CI of $[6.0, 45.2]$ ms, confirming that global latency thresholds are artifacts of operational pooling across heterogeneous machines.

pairwise correlations demonstrate no linear dependence between network telemetry and manufacturing features.

### B. Temporal Stationarity and Granger Causality (RQ2)

Application of the Augmented Dickey-Fuller test to the continuous network latency and production speed time-series yields ADF statistics of $t = -14.49$ ($p < 0.001$) and $t = -14.36$ ($p < 0.001$), confirming weak stationarity.

Table IV presents the Granger causality test summary evaluating whether past network latency predicts current production speed across all 450 machine-mode-lag configurations from `reports/granger_causality_results.csv`.

Across all 450 tested configurations, exactly 23 nominally pass (5.1%, matching the nominal false-positive rate under the null). Following Benjamini-Hochberg adjustment, exactly zero (0) tests demonstrate statistically significant temporal precedence. Latency fluctuations do not Granger-cause production speed variations.

### C. Structural Breakpoint Analysis: Naive $F$ vs. Permutation Test (RQ3)

Evaluating segmented linear regression models across discrete operational modes identifies candidate breakpoints in latency. In Active mode, the exhaustive search algorithm selects a threshold candidate at $\hat{\tau} = 20.69$ ms ($\approx 20.7$ ms). Table V contrasts the naive asymptotic $F$-test against the supremum-$F$ permutation test ($N = 2,000$ iterations, seed=42, archived in `reports/breakpoint_permutation_tests.csv`).

While the naive $F$-test reports an apparently significant $p = 0.0378$ ($F = 4.3127$) for Active mode, the exact

TABLE V
STRUCTURAL BREAKPOINT EVALUATION IN NETWORK LATENCY VS. PRODUCTION SPEED

| Data Slice | Candidate $\tau$ | Sup-$F$ | Naive $p$ | Perm. $p$ | FDR $q$ |
|---|---|---|---|---|---|
| Pooled | 41.22 ms | 1.0532 | 0.3048 | 0.7530 | 0.7530 |
| Active | 20.69 ms | 4.3127 | 0.0378 | 0.1640 | 0.5100 |
| Idle | 27.55 ms | 1.2800 | 0.2579 | 0.6890 | 0.7530 |
| Maintenance | 25.50 ms | 3.2382 | 0.0720 | 0.2550 | 0.5100 |

TABLE VI
MULTI-CLUSTER K-MEANS EVALUATION ON STANDARDIZED NETWORK TELEMETRY ($k = 2..6$)

| Clusters $k$ | Inertia | Silhouette† | Davies-Bouldin | Calinski-Harabasz |
|---|---|---|---|---|
| $k = 2$ | 125,071 | 0.3548 | 1.1871 | 59,910 |
| $k = 3$ | 79,315 | 0.3764 | 0.8687 | 76,078 |
| **$k = 4$** | **50,230** | **0.4063** | **0.7690** | **99,386** |
| $k = 5$ | 42,314 | 0.3908 | 0.8367 | 93,161 |
| $k = 6$ | 35,980 | 0.3745 | 0.8422 | 91,169 |

†Silhouette scores evaluated on a representative random subsample of $n = 15,000$ observations (seed 42 in `src/network_profiling.py`) due to $\mathcal{O}(N^2)$ pairwise distance complexity; Inertia, Davies-Bouldin, and Calinski-Harabasz indices evaluate all $N = 100,000$ points.

supremum permutation test yields $p_{\text{perm}} = 0.1640$, corresponding to a Benjamini-Hochberg false discovery rate of $q = 0.5100$. Under the permuted null distribution where latency and production speed are independent, a maximal $F$-statistic of 4.3127 or greater arises by pure chance in 16.4% of random realizations.

Crucially, under the permutation null, the naive test rejects the null at nominal $\alpha = 0.05$ in 20.45% of realizations (409 out of 2,000 permutations), demonstrating a more than fourfold inflation of the type-I error rate when using standard asymptotic tables for data-chosen breakpoints. The 20.7 ms threshold is a classic artifact of data-dependent threshold selection and cannot be cited as a confirmed operational guardrail.

### D. Unsupervised Network Profiling (RQ4)

Evaluating $K$-means clustering across candidate cluster counts $k \in [2, 6]$ on standardized network features (`StandardScaler`) produces the selection ladder summarized in Table VI (archived in `reports/clustering_k_selection.csv`).

The partition at $k = 4$ maximizes the silhouette score (0.4063) [25] and minimizes the Davies-Bouldin index (0.7690) [26]. However, silhouette scores below 0.50 are consistent with an arbitrary geometric Voronoi partition of a continuous uniform bivariate plane rather than discrete physical operational regimes. Similarly, Calinski-Harabasz metrics (99,386) [27] scale directly with sample size ($N = 100,000$), reflecting geometric density rather than clustering quality.

### E. Supervised Predictive Modeling Ladder (RQ5)

Table VII presents the empirical performance ladder across model formulations and validation splits from

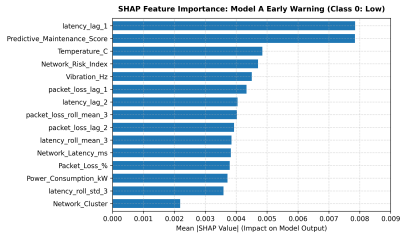



Fig. 3. SHAP feature attribution summary for Model A (Early Warning, Horizon B). Nominal top feature is latency_lag_1, but all mean absolute SHAP values satisfy $|\text{SHAP}| \leq 0.008$ across all features, confirming that feature importance rankings are at noise level and yield near-chance discrimination.

`reports/model_baseline_ladder.csv`, with bootstrap 95% confidence intervals for ROC-AUC and class-specific precision and recall for the minority High-efficiency class.

The empirical findings are consistent across all architectures: (1) **Near-Chance Predictive Discrimination**: Across all configurations, ROC-AUC values remain strictly confined between $0.4738$ and $0.5073$, with bootstrap 95% confidence intervals overlapping $0.5000$. Tree-based models cannot reliably separate operational efficiency classes using network telemetry. (2) **Macro F1 Ceiling**: Random Forest achieves a nominal Macro F1 of $0.3368$ on Horizon B, which does not represent meaningful predictive capture of minority classes (High efficiency recall is only $3.05\%$). (3) **Generalization on Unseen Machines**: Comparing like with like under Horizon B, Random Forest achieves $0.3368$ Macro F1 on seen machines versus $0.3275$ on unseen machines. Both operate near chance, confirming that performance reflects uninformative baselines rather than authentic model generalization.

## V. Discussion

**A. Answers to Research Questions**—We systematically address the five research questions: **RQ1 (Distribution)**: Formal KS tests show no evidence against continuous uniformity for all nine telemetry variables after Bonferroni correction (all-nine Bonferroni bound $p_{\text{adj}} = 0.1947$). Pairwise correlations demonstrate no linear dependence ($|r| \leq 0.0075$, none survive Bonferroni across 36 pairs); telemetry is statistically consistent with independent uniform generation. **RQ2 (Causality)**: Telemetry streams are weakly stationary ($p < 0.001$), but Granger causality tests across 450 machine-mode-lag configurations reveal zero statistically significant predictive precedence after FDR control ($0/450$ significant). **RQ3 (Breakpoint)**: The apparent 20.7 ms latency breakpoint is an artifact of candidate quantile selection ($F = 4.3127$, naive asymptotic $p = 0.0378$). A supremum-$F$ permutation test reveals $p = 0.1640$ (FDR $q = 0.5100$), confirming the breakpoint is exploratory. **RQ4 (Regimes)**: Multi-$k$ clustering demonstrates that $k = 4$ maximizes the silhouette score ($0.4063$ under `StandardScaler`), reflecting an arbitrary geometric partitioning of a continuous uniform bivariate space rather than discrete physical states. **RQ5 (Prediction)**: Under leak-free temporal and grouped splitting, network telemetry achieves near-chance classification (ROC-AUC 0.4738–0.5073), failing to beat random guessing when direct label ingredients are excluded.

**B. Methodological Hazard of Data-Chosen Breakpoints**—Our empirical demonstration contrasting the naive $F$-test ($p = 0.0378$) against the permutation test ($p = 0.1640$) illustrates a fundamental methodological lesson for industrial informatics. In sensor telemetry, scanning for operational degradation thresholds across continuous grids is common practice. When the threshold is chosen to maximize fit, standard asymptotic $F$-distributions no longer apply. Under the permutation null, the naive test yields a false positive rate of $20.45\%$. Without permutation resampling or Hansen's sample-splitting corrections [22], researchers risk publishing spurious thresholds that fail to replicate in live manufacturing environments.

**C. The Continuum Partitioning Fallacy**—A parallel hazard occurs in unsupervised network profiling. Applying clustering algorithms (e.g., $K$-means) to uniform or uncoupled bivariate telemetry will always yield mathematical partitions. When Calinski-Harabasz metrics scale with large sample sizes ($N = 100,000$), high values are consistent with sample size scaling rather than cluster separability, and analysts may mistake these sectors for authentic network Quality of Service (QoS) classes. Rigorous validation against null uniform reference models is mandatory.

**D. Alignment with 3GPP TS 22.104 and Operational Reality**—Standards such as 3GPP TS 22.104 [5] mandate strict end-to-end latency and jitter bounds across motion control scenarios (Clause 5.1.1). In authentic factory environments, exceeding these limits causes PLC buffer overflows, emergency packet drops, and physical machine stops. In the analyzed benchmark, however, latency fluctuates uniformly between 1 ms and 50 ms without impacting machine physics. This highlights the urgent need for authentic open-access OT/IT datasets captured from physical testbeds.

**E. Threats to Validity and Limitations**—The primary limitation of this case study is that the underlying dataset is synthetic. While the benchmark provides realistic enterprise schemas, its uniform feature distributions prevent the emergence of genuine cyber-physical feedback loops. Our analysis does not imply that 6G URLLC is irrelevant to smart factories; rather, it proves that the benchmark dataset cannot support causal inferences regarding network QoS.

## VI. Conclusion and Future Work

This paper presented an end-to-end, leakage-safe temporal-evidence pipeline for smart factory telemetry. Through rigorous statistical audits, we established that the benchmark dataset shows no evidence against independent uniform generation, disconfirmed a naive 20.7 ms latency breakpoint via supremum-$F$ permutation testing ($p = 0.1640$), and demonstrated that network telemetry alone yields near-chance predictive power (ROC-AUC 0.4738–0.5073) under leak-free validation. Our findings underscore the critical importance of auditing generative distributions, guarding against proxy feature leakage, and validating data-chosen changepoints. Future work

TABLE VII
SUPERVISED MODEL PERFORMANCE LADDER ACROSS CHRONOLOGICAL AND GROUPED SPLITS

| Model Formulation | Algorithm | Acc. | Macro F1 | Wtd F1 | ROC-AUC [95% CI] | Hi-P | Hi-R | Operational Interpretation |
|---|---|---|---|---|---|---|---|---|
| **Model A (Horizon B: 30-min)** | Dummy Majority | 0.7842 | 0.2930 | 0.6894 | 0.5000 [0.5000, 0.5000] | 0.0000 | 0.0000 | Naive majority class baseline |
| | Logistic Regression | 0.2791 | 0.2321 | 0.3554 | 0.5029 [0.4828, 0.5253] | 0.0302 | 0.4656 | Underperforms majority baseline |
| | Random Forest | 0.6730 | 0.3368 | 0.6622 | 0.5068 [0.4856, 0.5296] | 0.0404 | 0.0305 | Near-chance discrimination |
| | LightGBM | 0.4313 | 0.2966 | 0.5014 | 0.5073 [0.4863, 0.5283] | 0.0295 | 0.2061 | Near-chance discrimination |
| **Model A (Horizon A: 1-shift)** | Random Forest | 0.5284 | 0.3126 | 0.5742 | 0.4949 [0.4851, 0.5048] | 0.0291 | 0.1117 | Near-chance discrimination |
| | LightGBM | 0.3725 | 0.2707 | 0.4470 | 0.4988 [0.4891, 0.5089] | 0.0315 | 0.2709 | Near-chance discrimination |
| **Model B (Diagnosis)** | Random Forest | 0.5569 | 0.3246 | 0.5947 | 0.5016 [0.4916, 0.5114] | 0.0240 | 0.0590 | Near-chance discrimination |
| | LightGBM | 0.3789 | 0.2733 | 0.4545 | 0.4994 [0.4895, 0.5078] | 0.0300 | 0.2574 | Near-chance discrimination |
| **Model A (Unseen Machines, Hor. B)** | Random Forest | 0.7024 | 0.3275 | 0.6717 | 0.4896 [0.4675, 0.5134] | 0.0141 | 0.0078 | No better than chance |
| | LightGBM | 0.4435 | 0.2889 | 0.5108 | 0.4738 [0.4539, 0.4948] | 0.0215 | 0.1328 | No better than chance |

Note: High denotes pristine operational efficiency ($\mathcal{Y} = \text{High}$, $2.986\%$ prevalence). Unseen Machines evaluates Horizon B ($H = 30$ min) on held-out Machines 41–50 ($N = 20,016$) comparing directly with seen machines under Horizon B. 95% confidence intervals for ROC-AUC are estimated via 1,000 bootstrap resamples on the test set.

will deploy this open-source pipeline on physical, hardware-in-the-loop 5G/6G industrial testbeds to capture authentic nonlinear network-physical interactions.

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